

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2197

Auburn 100 kV
August 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2197. The Interconnection Customer proposes a 78.9 MW solar plus storage facility interconnecting at the Auburn 100 kV substation in the Indiana area. The request was received on October 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 745 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,912,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Roseau County to Barret 100kV	Monte Del Estado 6303 to Harper 100kV L	164.9	234	7.0
Hassen Cassel to Darlington Rd 100kV	Wayne Trace to Darlington Rd 100kV	147.8	189	5.1
Muldoon Mill to Hassen Cassel 100kV	Darlington Road to Wayne Trace 100kV line	143.6	204	5.1
Roseau County to South Whitehall 100kV	Monte Del Estado 6303 to Harper 100kV	142.5	172	7.1
Lake Marion to Point A 100kV	State St to Robison Park 230kV	127.8	224	5.0
Barret to Point A 100kV	Robison Park to State St 230kV	124.7	247	5.0
Hillcrest to Muldoon Mill Substation 100kV	Wayne Trace to Darlington Road 100kV	114.5	435	8.1
Harper to Monte Del Estad 6303 100kV L	Roseau County to Barret 100kV	108.0	423	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Roseau County to Barret 100kV	Rebuild line with higher-capacity conductor	\$8,131,000	25% (\$2,017,000)
Hassen Cassel to Darlington Rd 100kV	Rebuild line with higher-capacity conductor	\$1,881,000	20% (\$379,000)
Hassen Cassel to Muldoon Mill 100kV	Rebuild line with higher-capacity conductor	\$2,211,000	20% (\$445,000)
South Whitehall to Roseau County Substation 100kV	Rebuild line with higher-capacity conductor	\$15,689,000	23% (\$3,596,000)
Lake Marion to Point A 100kV	Rebuild line with higher-capacity conductor	\$5,863,000	38% (\$2,201,000)
Barret to Point A 100kV	Rebuild line with higher-capacity conductor	\$24,228,000	38% (\$9,094,000)
Hillcrest to Muldoon Mill Substation 100kV	Rebuild line with higher-capacity conductor	\$6,079,000	2% (\$125,000)
Harper to Monte Del Estad 6303 100kV L	Reconductor line	\$2,495,000	2% (\$55,000)
	Total allocated to project		\$17,912,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.